SCR2043 OPERATING SYSTEMS

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Section	: 03	

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. **Download necessary source codes:**

   Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. **Compile the source files:**

   Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. **Execute the dummy processes:**

   Run all the dummy processes

```
./mainprocess &
```

   Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command
`ps -e`
Output

```
    PID TTY          TIME CMD
      1 ?        00:00:04 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 pool_workqueue_release
      4 ?        00:00:00 kworker/R-rcu_g
      5 ?        00:00:00 kworker/R-rcu_p
      6 ?        00:00:00 kworker/R-slub_
      7 ?        00:00:00 kworker/R-netns
     10 ?        00:00:00 kworker/0:0H-kblockd
     11 ?        00:00:00 kworker/u4:0-ext4-rsv-conversion
     12 ?        00:00:00 kworker/R-mm_pe
     13 ?        00:00:00 rcu_tasks_kthread
     14 ?        00:00:00 rcu_tasks_rude_kthread
     15 ?        00:00:00 rcu_tasks_trace_kthread
     16 ?        00:00:00 ksoftirqd/0
     17 ?        00:00:00 rcu_preempt
     18 ?        00:00:00 migration/0
     19 ?        00:00:00 idle_inject/0
     20 ?        00:00:00 cpuhp/0
     21 ?        00:00:00 cpuhp/1
     22 ?        00:00:00 idle_inject/1
     23 ?        00:00:00 migration/1
     24 ?        00:00:00 ksoftirqd/1
     26 ?        00:00:00 kworker/1:0H-events_highpri
     27 ?        00:00:00 kworker/u5:0-events_power_efficient
     29 ?        00:00:00 kdevtmpfs
     30 ?        00:00:00 kworker/R-inet_
     31 ?        00:00:00 kworker/u5:1-events_power_efficient
     32 ?        00:00:00 kauditd
     33 ?        00:00:00 khungtaskd
     34 ?        00:00:00 oom_reaper
     35 ?        00:00:00 kworker/u5:2-events_unbound
     36 ?        00:00:00 kworker/R-write
     37 ?        00:00:00 kcompactd0
     38 ?        00:00:00 ksmd
     39 ?        00:00:00 kworker/1:1-cgwb_release
     40 ?        00:00:00 khugepaged
     41 ?        00:00:00 kworker/R-kinte
     42 ?        00:00:00 kworker/R-kbloc
     43 ?        00:00:00 kworker/R-blkcg
     44 ?        00:00:00 irq/9-acpi
```

```
     45 ?        00:00:00 kworker/R-tpm_d
     46 ?        00:00:00 kworker/R-ata_s
     47 ?        00:00:00 kworker/R-md
     48 ?        00:00:00 kworker/R-md_bi
     49 ?        00:00:00 kworker/R-edac-
     50 ?        00:00:00 kworker/R-devfr
     51 ?        00:00:00 watchdogd
     52 ?        00:00:00 kworker/u6:1-events_unbound
     53 ?        00:00:00 kworker/1:1H-kblockd
     54 ?        00:00:00 kswapd0
     55 ?        00:00:00 ecryptfs-kthread
     56 ?        00:00:00 kworker/R-kthro
     57 ?        00:00:00 kworker/R-acpi_
     58 ?        00:00:00 scsi_eh_0
     59 ?        00:00:00 kworker/R-scsi_
     60 ?        00:00:00 scsi_eh_1
     61 ?        00:00:00 kworker/R-scsi_
     62 ?        00:00:00 kworker/0:1H-kblockd
     63 ?        00:00:00 kworker/R-mld
     64 ?        00:00:00 kworker/R-ipv6_
     72 ?        00:00:00 kworker/R-kstrp
     74 ?        00:00:00 kworker/u7:0
     75 ?        00:00:00 kworker/u8:0
     76 ?        00:00:00 kworker/u9:0
     82 ?        00:00:00 kworker/R-crypt
     83 ?        00:00:00 kworker/0:2-events
     92 ?        00:00:00 kworker/R-charg
    133 ?        00:00:03 kworker/0:3-events
    134 ?        00:00:00 kworker/u4:1
    136 ?        00:00:00 kworker/R-mpt_p
    137 ?        00:00:00 kworker/R-mpt/0
    156 ?        00:00:00 scsi_eh_2
    157 ?        00:00:00 kworker/R-scsi_
    191 ?        00:00:00 kworker/R-raid5
    228 ?        00:00:00 jbd2/sda1-8
    229 ?        00:00:00 kworker/R-ext4-
    296 ?        00:00:00 psimon
    301 ?        00:00:00 systemd-journal
    322 ?        00:00:00 kworker/R-kmpat
    323 ?        00:00:00 kworker/R-kmpat
    324 ?        00:00:00 kworker/u6:3-events_power_efficient
    359 ?        00:00:00 multipathd
    370 ?        00:00:00 systemd-udevd
    371 ?        00:00:01 kworker/1:4-events
    379 ?        00:00:00 psimon
    455 ?        00:00:00 irq/18-vmwgfx
    456 ?        00:00:00 kworker/R-ttm
    478 ?        00:00:00 jbd2/sda16-8
    479 ?        00:00:00 kworker/R-ext4-
    511 ?        00:00:00 systemd-network
    528 ?        00:00:00 systemd-resolve
    533 ?        00:00:00 systemd-timesyn
    636 ?        00:00:00 dbus-daemon
    641 ?        00:00:00 polkitd
    651 ?        00:00:00 systemd-logind
    657 ?        00:00:00 udisksd
    702 ?        00:00:00 unattended-upgr
    707 ?        00:00:00 rsyslogd
    735 ?        00:00:00 ModemManager
```

```
    775 ?        00:00:00 cron
    804 tty1     00:00:00 agetty
    865 ?        00:00:00 sshd
    906 ?        00:00:00 sshd
    908 ?        00:00:01 sshd
    909 pts/0    00:00:00 bash
    951 pts/0    00:00:00 mainprocess
    952 pts/0    00:00:00 mainprocess
    953 pts/0    00:00:00 mainprocess
    954 pts/0    00:00:00 subprocess2
    955 pts/0    00:00:00 subprocess2
    956 pts/0    00:00:00 subprocess2
    957 pts/0    00:00:00 subprocess1
    958 pts/0    00:00:00 subprocess1
    964 ?        00:00:00 kworker/u6:0-flush-8:0
    965 ?        00:00:00 kworker/1:0
    966 pts/0    00:00:00 ps
```

Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command
`ps -ef`
Output

```
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 02:43 ?        00:00:04 /sbin/init
root           2       0  0 02:43 ?        00:00:00 [kthreadd]
root           3       2  0 02:43 ?        00:00:00 [pool_workqu
root           4       2  0 02:43 ?        00:00:00 [kworker/R-r
root           5       2  0 02:43 ?        00:00:00 [kworker/R-r
root           6       2  0 02:43 ?        00:00:00 [kworker/R-s
root           7       2  0 02:43 ?        00:00:00 [kworker/R-n
root          10       2  0 02:43 ?        00:00:00 [kworker/0:0
root          11       2  0 02:43 ?        00:00:00 [kworker/u4:
root          12       2  0 02:43 ?        00:00:00 [kworker/R-m
root          13       2  0 02:43 ?        00:00:00 [rcu_tasks_k
root          14       2  0 02:43 ?        00:00:00 [rcu_tasks_r
root          15       2  0 02:43 ?        00:00:00 [rcu_tasks_t
root          16       2  0 02:43 ?        00:00:00 [ksoftirqd/0
root          17       2  0 02:43 ?        00:00:00 [rcu_preempt
root          18       2  0 02:43 ?        00:00:00 [migration/0
root          19       2  0 02:43 ?        00:00:00 [idle_inject
root          20       2  0 02:43 ?        00:00:00 [cpuhp/0]
root          21       2  0 02:43 ?        00:00:00 [cpuhp/1]
root          22       2  0 02:43 ?        00:00:00 [idle_inject
root          23       2  0 02:43 ?        00:00:00 [migration/1
root          24       2  0 02:43 ?        00:00:00 [ksoftirqd/1
root          26       2  0 02:43 ?        00:00:00 [kworker/1:0
root          27       2  0 02:43 ?        00:00:00 [kworker/u5:
root          29       2  0 02:43 ?        00:00:00 [kdevtmpfs]
root          30       2  0 02:43 ?        00:00:00 [kworker/R-i
root          31       2  0 02:43 ?        00:00:00 [kworker/u5:
root          32       2  0 02:43 ?        00:00:00 [kauditd]
root          33       2  0 02:43 ?        00:00:00 [khungtaskd]
root          34       2  0 02:43 ?        00:00:00 [oom_reaper]
```

```
root          35       2  0 02:43 ?        00:00:00 [kworker/u5:
root          36       2  0 02:43 ?        00:00:00 [kworker/R-w
root          37       2  0 02:43 ?        00:00:00 [kcompactd0]
root          38       2  0 02:43 ?        00:00:00 [ksmd]
root          39       2  0 02:43 ?        00:00:00 [kworker/1:1
root          40       2  0 02:43 ?        00:00:00 [khugepaged]
root          41       2  0 02:43 ?        00:00:00 [kworker/R-k
root          42       2  0 02:43 ?        00:00:00 [kworker/R-k
root          43       2  0 02:43 ?        00:00:00 [kworker/R-b
root          44       2  0 02:43 ?        00:00:00 [irq/9-acpi]
root          45       2  0 02:43 ?        00:00:00 [kworker/R-t
root          46       2  0 02:43 ?        00:00:00 [kworker/R-a
root          47       2  0 02:43 ?        00:00:00 [kworker/R-m
root          48       2  0 02:43 ?        00:00:00 [kworker/R-m
root          49       2  0 02:43 ?        00:00:00 [kworker/R-e
root          50       2  0 02:43 ?        00:00:00 [kworker/R-d
root          51       2  0 02:43 ?        00:00:00 [watchdogd]
root          52       2  0 02:43 ?        00:00:00 [kworker/u6:
root          53       2  0 02:43 ?        00:00:00 [kworker/1:1
root          54       2  0 02:43 ?        00:00:00 [kswapd0]
root          55       2  0 02:43 ?        00:00:00 [ecryptfs-kt
root          56       2  0 02:43 ?        00:00:00 [kworker/R-k
root          57       2  0 02:43 ?        00:00:00 [kworker/R-a
root          58       2  0 02:43 ?        00:00:00 [scsi_eh_0]
root          59       2  0 02:43 ?        00:00:00 [kworker/R-s
root          60       2  0 02:43 ?        00:00:00 [scsi_eh_1]
root          61       2  0 02:43 ?        00:00:00 [kworker/R-s
root          62       2  0 02:43 ?        00:00:00 [kworker/0:1
root          63       2  0 02:43 ?        00:00:00 [kworker/R-m
root          64       2  0 02:43 ?        00:00:00 [kworker/R-i
root          72       2  0 02:43 ?        00:00:00 [kworker/R-k
root          74       2  0 02:43 ?        00:00:00 [kworker/u7:
root          75       2  0 02:43 ?        00:00:00 [kworker/u8:
root          76       2  0 02:43 ?        00:00:00 [kworker/u9:
root          82       2  0 02:43 ?        00:00:00 [kworker/R-c
root          83       2  0 02:43 ?        00:00:00 [kworker/0:2
root          92       2  0 02:43 ?        00:00:00 [kworker/R-c
root         133       2  0 02:43 ?        00:00:04 [kworker/0:3
root         134       2  0 02:43 ?        00:00:00 [kworker/u4:
root         136       2  0 02:43 ?        00:00:00 [kworker/R-m
root         137       2  0 02:43 ?        00:00:00 [kworker/R-m
root         156       2  0 02:43 ?        00:00:00 [scsi_eh_2]
root         157       2  0 02:43 ?        00:00:00 [kworker/R-s
root         191       2  0 02:43 ?        00:00:00 [kworker/R-r
root         228       2  0 02:43 ?        00:00:00 [jbd2/sda1-8
root         229       2  0 02:43 ?        00:00:00 [kworker/R-e
root         296       2  0 02:43 ?        00:00:00 [psimon]
root         301       1  0 02:43 ?        00:00:00 /usr/lib/sys
root         322       2  0 02:43 ?        00:00:00 [kworker/R-k
root         323       2  0 02:43 ?        00:00:00 [kworker/R-k
root         324       2  0 02:43 ?        00:00:00 [kworker/u6:
root         359       1  0 02:43 ?        00:00:00 /sbin/multip
root         370       1  0 02:43 ?        00:00:00 /usr/lib/sys
root         371       2  0 02:43 ?        00:00:01 [kworker/1:4
root         379       2  0 02:43 ?        00:00:00 [psimon]
root         455       2  0 02:43 ?        00:00:00 [irq/18-vmwg
root         456       2  0 02:43 ?        00:00:00 [kworker/R-t
root         478       2  0 02:43 ?        00:00:00 [jbd2/sda16-
root         479       2  0 02:43 ?        00:00:00 [kworker/R-e
```

```
systemd+     511       1  0 02:43 ?        00:00:00 /usr/lib/sys
systemd+     528       1  0 02:43 ?        00:00:00 /usr/lib/sys
systemd+     533       1  0 02:43 ?        00:00:00 /usr/lib/sys
message+     636       1  0 02:43 ?        00:00:00 @dbus-daemon
polkitd      641       1  0 02:43 ?        00:00:00 /usr/lib/pol
root         651       1  0 02:43 ?        00:00:00 /usr/lib/sys
root         657       1  0 02:43 ?        00:00:00 /usr/libexec
root         702       1  0 02:43 ?        00:00:00 /usr/bin/pyt
syslog       707       1  0 02:43 ?        00:00:00 /usr/sbin/rs
root         735       1  0 02:43 ?        00:00:00 /usr/sbin/Mo
root         775       1  0 02:43 ?        00:00:00 /usr/sbin/cr
root         804       1  0 02:43 tty1     00:00:00 /sbin/agetty
root         865       1  0 02:44 ?        00:00:00 sshd: /usr/s
root         906     865  0 02:45 ?        00:00:00 sshd: lam [p
lam          908     906  0 02:46 ?        00:00:01 sshd: lam@pt
lam          909     908  0 02:46 pts/0    00:00:00 -bash
lam          951     909  0 02:47 pts/0    00:00:00 ./mainproces
lam          952     951  0 02:47 pts/0    00:00:00 ./mainproces
lam          953     951  0 02:47 pts/0    00:00:00 ./mainproces
lam          954     953  0 02:47 pts/0    00:00:00 ./subprocess
lam          955     953  0 02:47 pts/0    00:00:00 ./subprocess
lam          956     953  0 02:47 pts/0    00:00:00 ./subprocess
lam          957     952  0 02:47 pts/0    00:00:00 ./subprocess
lam          958     952  0 02:47 pts/0    00:00:00 ./subprocess
root         964       2  0 02:49 ?        00:00:00 [kworker/u6:
root         965       2  0 02:49 ?        00:00:00 [kworker/1:0
root         969       2  0 02:50 ?        00:00:00 [kworker/u5:
lam          970     909 99 02:52 pts/0    00:00:00 ps -ef
```

Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command
`ps -ef | grep '[s]ubprocess'`
Output

```
lam@secr2043:~$ ps -ef | grep '[s]ubprocess'
lam          954     953  0 02:47 pts/0    00:00:00 ./subprocess2
lam          955     953  0 02:47 pts/0    00:00:00 ./subprocess2
lam          956     953  0 02:47 pts/0    00:00:00 ./subprocess2
lam          957     952  0 02:47 pts/0    00:00:00 ./subprocess1
lam          958     952  0 02:47 pts/0    00:00:00 ./subprocess1
```

Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)

- Command (`cmd`)

Command
`ps -eo pid,user,pcpu,pmem,cmd | grep '[s]ubprocess'`
Output

```
lam@secr2043:~$ ps -eo pid,user,pcpu,pmem,cmd | grep '[s]ubprocess'
    954 lam         0.0  0.1 ./subprocess2
    955 lam         0.0  0.1 ./subprocess2
    956 lam         0.0  0.1 ./subprocess2
    957 lam         0.0  0.1 ./subprocess1
    958 lam         0.0  0.1 ./subprocess1
```

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (`pmem`)?

Command
`ps -eo pid,user,pcpu,pmem,cmd --sort=pmem| grep '[s]ubprocess'`
Output

```
lam@secr2043:~$ ps -eo pid,user,pcpu,pmem,cmd --sort=pmem| grep '[s]ubprocess'
    954 lam         0.0  0.1 ./subprocess2
    955 lam         0.0  0.1 ./subprocess2
    956 lam         0.0  0.1 ./subprocess2
    957 lam         0.0  0.1 ./subprocess1
    958 lam         0.0  0.1 ./subprocess1
```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command
`ps -eo pid,ppid,user,cmd --forest | grep -E 'mainprocess|subprocess1|subprocess2'`
Output

```
lam@secr2043:~$  ps -eo pid,ppid,user,cmd --forest | grep -E 'mainprocess|subprocess1|subprocess2'
    951     909 lam                  \_ ./mainprocess
    952     951 lam                  |   \_ ./mainprocess
    957     952 lam                  |   |   \_ ./subprocess1
    958     952 lam                  |   |   \_ ./subprocess1
    953     951 lam                  |   \_ ./mainprocess
    954     953 lam                  |       \_ ./subprocess2
    955     953 lam                  |       \_ ./subprocess2
    956     953 lam                  |       \_ ./subprocess2
   1014     909 lam                  \_ grep --color=auto -E mainprocess|subprocess1|subprocess2
```

Question 7

Use `pstree` command with option that show the number of threads to each process.

Command
`pstree -T`
Output

```
lam@secr2043:~$ pstree -T
systemd-+-ModemManager
        |-agetty
        |-cron
        |-dbus-daemon
        |-multipathd
        |-polkitd
        |-rsyslogd
        |-sshd---sshd---sshd---bash-+-mainprocess-+-mainprocess---2*[subprocess1]
        |                           |             `-mainprocess---3*[subprocess2]
        |                           `-pstree
        |-systemd-journal
        |-systemd-logind
        |-systemd-network
        |-systemd-resolve
        |-systemd-timesyn
        |-systemd-udevd
        |-udisksd
        `-unattended-upgr
lam@secr2043:~$ pstree -c -s 951
systemd---sshd---sshd---sshd---bash---mainprocess-+-mainprocess-+-subprocess1
                                                  |             `-subprocess1
                                                  `-mainprocess-+-subprocess2
                                                                |-subprocess2
                                                                `-subprocess2
```

Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

Command
`sudo renice -n -10 -p 957`
Output

```
lam@secr2043:~$ sudo renice -n -10 -p 957
[sudo] password for lam:
957 (process ID) old priority 0, new priority -10
```

Question 9

Terminate all running processes with the name “mainprocess”.

Command
`killall -15 mainprocess`
Output

```
lam@secr2043:~$ killall -15 mainprocess
Main process (ID: 952) received signal: 15. Terminating...
Main process (ID: 951) received signal: 15. Terminating...
```

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <unistd.h>

int main(){
        pid_t pid;

        /*fork a child process */
        pid = fork();

        if (pid < 0) { /* error occurred */
                fprintf(stderr, "Fork Failed");
                return 1;
        }
        else if (pid == 0) { /*child process */
                printf("Child Process. PID = %d\n", getpid());
                execlp("/bin/ls", "ls",NULL);
        }
        else { /* parent process */
                printf("Parent Process. PID = %d\n", getpid());
                /* parent will wait for the child to complete */
                wait(NULL);
                printf("Child Complete\n");
        }
        return 0;
}
```

```
 GNU nano 7.2                multiprocessing.c *
#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <unistd.h>

int main(){
        pid_t pid;

        /*fork a child process */
        pid = fork();

        if (pid < 0) { /* error occurred */
                fprintf(stderr, "Fork Failed");
                return 1;
        }
        else if (pid == 0) { /*child process */
                printf("Child Process. PID = %d\n", getpid());
                execlp("/bin/ls", "ls",NULL);
        }
        else { /* parent process */
                printf("Parent Process. PID = %d\n", getpid());
                /* parent will wait for the child to complete */
                wait(NULL);
                printf("Child Complete\n");
        }
        return 0;
}
```

Output:

```
lam@secr2043:~$ nano multiprocessing.c
lam@secr2043:~$ gcc multiprocessing.c -o multiprocessing
lam@secr2043:~$ ./multiprocessing
Parent Process. PID = 875
Child Process. PID = 876
activity3     mainprocess        nano.959.save  subprocess2
bye.txt       mainprocess.c      os_lab         subprocess2.c
example.txt   multiprocessing    share
greeting.txt  multiprocessing.c  subprocess1
hello.txt     myfile.txt         subprocess1.c
Child Complete
```